# Augmented Reality Application for Optimal Placement of Photovoltaic Inverters

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***Abstract*—This paper presents the development and validation of an augmented reality (AR) prototype application designed to support the technical inspection and optimal positioning of photovoltaic (PV) inverters. Motivated by the recurrent issue of rework and installation errors observed in solar engineering projects, this study integrates Building Information Modeling (BIM), parametric modeling, and mobile AR technologies to reduce uncertainty during site inspections. The prototype was developed using Revit for parametric equipment modeling, Unity with Vuforia for AR integration, and validated with seven users experienced in solar system installation and design. Results indicate high usability (PSSUQ overall score of 1.66) and strong fidelity between virtual and real-world distances, with an average measurement deviation of only 0.5 cm. The average inspection execution time was 4 minutes and 19 seconds, demonstrating a low learning curve and operational feasibility. The proposed AR tool proved viable for supporting decision-making during inverter placement and holds potential for broader applications in electrical, architectural, and civil engineering domains.**

***Index Terms*—Augmented Reality, Photovoltaic Systems, BIM, Solar Energy, Mobile Applications, Technical Inspection.**

## I. Introduction

Brazil's electricity matrix has historically relied on hydroelectric generation, which exceeded 90% of national production in the early 2000s [1]. Although hydropower supported decades of stable development, the 2001 energy crisis exposed the vulnerability of depending almost exclusively on a single source. This event triggered long-term efforts to diversify the energy matrix, and solar photovoltaic (PV) generation gradually became one of the most relevant alternatives. The regulatory advancements introduced by ANEEL Resolution 482/2012 [4], combined with state-level tax incentives and progressive reductions in equipment costs, created favorable conditions for the expansion of distributed-generation systems [5], [6]. As a result, PV installations have grown rapidly, with Brazil reaching more than 14 GW of distributed PV capacity in 2022.

At the same time, the fourth industrial revolution, commonly associated with Industry 4.0, fostered the incorporation of technologies such as cyber-physical systems, artificial intelligence, advanced automation, BIM, and immersive environments. These technologies have played an essential role in modernizing engineering workflows by enabling real-time data processing, improved visualization, and increased precision during technical decision-making. In the Architecture, Engineering, and Construction sector, Building Information Modeling (BIM) has become a central methodology that replaces traditional 2D drafting with information-rich parametric models. BIM allows design teams to anticipate conflicts, reduce inconsistencies, and coordinate project information throughout the entire lifecycle of a facility [7]–[9].

When combined with augmented reality (AR), BIM expands its capabilities by allowing virtual elements to be visualized directly within physical environments. This integration enhances spatial understanding and supports critical decisions during field activities. Prior work has shown that AR can improve geometric perception, facilitate inspection processes, and provide faster comprehension of design constraints [22], [25], [28]. In addition, AR tools developed for the solar energy sector have demonstrated promising results in tasks related to irradiation assessment, remote monitoring, and operator training [30], [31].

Despite these technological advances, PV installation workflows continue to face challenges related to incorrect measurements, unclear communication between field teams and engineering staff, and inconsistencies in documentation. In the company investigated in this study, 4% of all engineering redesigns were directly associated with inaccurate inverter placement or incomplete field measurements. These errors resulted in project delays, additional material consumption, increased labor hours, and unnecessary costs.

Motivated by this challenge, this work proposes an AR-based solution to enhance the accuracy and clarity of PV inverter placement during site inspections. The prototype integrates BIM-based parametric modeling with real-time AR visualization to ensure compliance with technical clearances, ventilation requirements, and spatial constraints.

## II. Related Work

Augmented reality research began to gain academic importance after Azuma's comprehensive conceptualization of AR as the integration of virtual elements within real environments through real-time interaction and spatial registration [22]. Since then, AR applications have expanded across numerous

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fields, including education, manufacturing, architecture, and engineering. In the AEC industry, AR improves understanding of spatial relationships and enables users to observe digital models superimposed on physical construction sites, which assists in design review, clash detection, and visual communication between project stakeholders [26], [28].

The integration of BIM and AR has been a recurring research topic, with studies highlighting advantages such as improved construction sequencing, enhanced visualization of building components, and more efficient identification of design conflicts. BIM provides structured information that is directly applicable to AR interfaces, allowing real-time inspection and more precise alignment between digital models and the physical environment [9], [36]. Research has also reported positive outcomes in educational and training contexts, where AR has been used to improve learners' understanding of complex spatial concepts and to reduce the cognitive load associated with traditional 2D documentation [25], [27].

Specific applications of AR in the solar energy sector have been explored as well. Previous studies investigated the use of AR for solar irradiance analysis, PV system performance assessment, and remote maintenance assistance [30], [31]. These works demonstrated that AR can support both technical and operational tasks in solar energy management by improving the accuracy of environmental assessments and facilitating quick diagnostic visualization.

Despite these contributions, there is limited research addressing AR-assisted inverter placement in distributed-generation PV systems. Existing studies focus primarily on solar panel positioning, shading analysis, or maintenance tasks, while inverter installation remains a critical activity that depends on accurate dimensional assessment and correct spatial interpretation of manufacturer requirements. Errors in inverter placement often arise from inadequate field measurements or ambiguous photographic documentation, which illustrates an important gap in the literature.

Usability evaluation techniques, such as the Post-Study System Usability Questionnaire (PSSUQ) [38], have been widely adopted in AR research to assess system effectiveness, information quality, and user satisfaction. Prior findings indicate that well-designed AR applications tend to achieve high usability, low learning curves, and strong acceptance in spatially oriented tasks [25], [29]. These characteristics reinforce the suitability of AR for inspection workflows that require precise spatial interpretation.

The present study contributes to this field by developing a BIM-integrated AR prototype tailored specifically to inverter installation in PV systems. The work also provides an empirical usability assessment and measurement accuracy analysis, supporting the viability of AR as a tool for reducing rework and improving field-to-office communication in solar engineering.

## III. Methodology

The methodology adopted in this study was structured to ensure that the development of the AR application was technically consistent, replicable, and aligned with real market needs in PV engineering. The process was divided into sequential stages that began with the identification of practical issues in inverter installation workflows and continued through requirements definition, BIM-based parametric modeling, AR system integration, and user validation. Each phase contributed specific insights that guided the refinement and assessment of the prototype.

### A. Market Applicability

The first stage involved analyzing the operational context of PV engineering teams to understand the origin of rework related to inverter installation. Field inspection reports, photographic documentation, and feedback from designers were examined to identify recurring issues in measurement accuracy and communication. The analysis revealed that 4% of the engineering redesigns were linked to incorrect inverter placement or incomplete field measurements. This diagnosis confirmed the potential for an AR-based solution to support spatial evaluation during site inspections.

### B. Prototype Requirements

Based on market observations, a set of functional and technical requirements was established. The AR prototype needed to run on Android devices, support real-scale visualization of inverter components, allow movement around virtual objects from a first-person perspective, and provide dimensional information regarding mandatory clearances. The system also required the ability to register images and videos for engineering documentation. These requirements served as the foundation for the system architecture and guided the selection of the software stack.

### C. Parametric Modeling in BIM

The inverter, AC and DC protection boxes, and cable trays were modeled using Autodesk Revit 2023. Parametric families were created in full scale to ensure that dimensions, geometric behaviors, and mounting constraints matched those of real equipment. The BIM approach facilitated the organization of technical information and enabled the seamless export of 3D models to AR development environments. The models were validated internally before integration, ensuring consistency in dimensions and metadata. Figure 1 shows the development and tests.

### D. AR Integration Workflow

The AR application was developed using Unity 3D, which provided a flexible environment for handling 3D assets, animations, interactions, and camera controls. Vuforia Engine was integrated with Unity to enable marker-based AR. A QR code was selected as the image target due to its robustness and ease of identification in various lighting conditions. The Revit-exported models were imported into Unity, where adjustments related to scale, orientation, pivot alignment, and visual fidelity were performed. The marker recognition workflow allowed the system to position the inverter model accurately with respect to the physical environment.

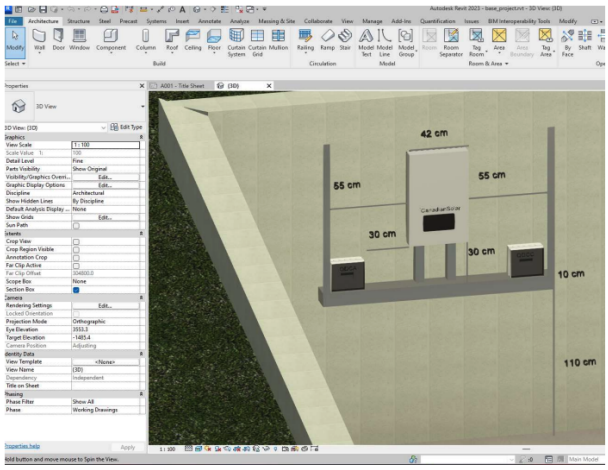


Fig. 1. Family testing in a 3D scenario in Revit

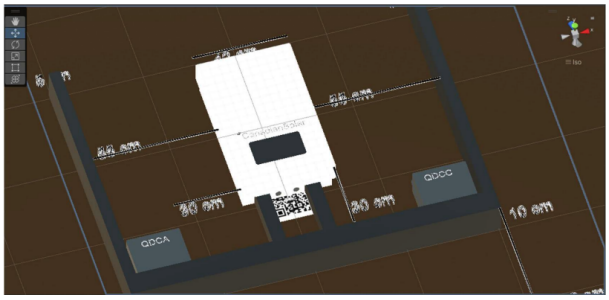


Fig. 2. Imported electrical equipment in Unity software

### E. Hardware Setup

Validation activities were conducted using a Samsung Galaxy S10e smartphone running Android 12. The device featured 6 GB of RAM, a 3,100 mAh battery, and sufficient processing power to render AR content smoothly. These hardware characteristics ensured that the results obtained in the validation stage represented realistic performance for typical field technicians who rely on mobile devices during inspections.

### F. User Validation

To evaluate the usability and accuracy of the prototype, a structured user test was implemented. Seven participants with experience in PV installation or design were selected. The validation consisted of a pre-use questionnaire, an operational script instructing users to execute tasks such as visualizing equipment, checking clearances, and capturing evidence, followed by the PSSUQ usability questionnaire. Time measurements and dimensional accuracy metrics were collected for each participant. These results were later analyzed to assess system usability, information clarity, and measurement fidelity.

## IV. Results and Discussion

This section presents the results obtained from the validation of the AR prototype, including analyzes of measurement accuracy, execution time, usability, perceived information quality, and qualitative feedback from participants. The results were collected from seven users with prior experience in PV installation or design. The evaluation followed a structured workflow composed of a pre-use questionnaire, a guided interaction script, field measurement exercises, and a post-use PSSUQ usability assessment.

### A. Measurement Accuracy Analysis

The prototype's ability to display clearance dimensions with high fidelity was evaluated by comparing distances provided by the AR application with real-world measurements obtained manually using a tape measure. Each participant validated eight clearances, yielding 56 total data points. The process of collect can be seen on Figure 3.

Fig. 3. Validation process of virtual object quotas in the real world.

The mean deviation between the AR-displayed values and the real measurements was approximately 0.5 cm, representing a highly accurate spatial correspondence. These result support the feasibility of applying AR for field dimensional verification.

### B. Execution Time Analysis

Execution time was collected from the moment users activated the AR view until the final evidence capture. The average completion time was 4 minutes and 19 seconds, demonstrating a low learning curve and indicating that the system is suitable for use during early-stage inspections.

To enrich the analysis, the time distribution across participants is shown in Figure 4 through a boxplot representation. The System Use category obtained the best performance, indicating that participants found the prototype intuitive and easy to interact with. The Information Quality category presented higher variance, mainly due to occasional instability of AR anchoring under windy outdoor conditions.

### C. Usability Assessment (PSSUQ)

The PSSUQ questionnaire included categories related to system usefulness (SysUse), information quality (InfoQual), and overall satisfaction. Table I presents the results.

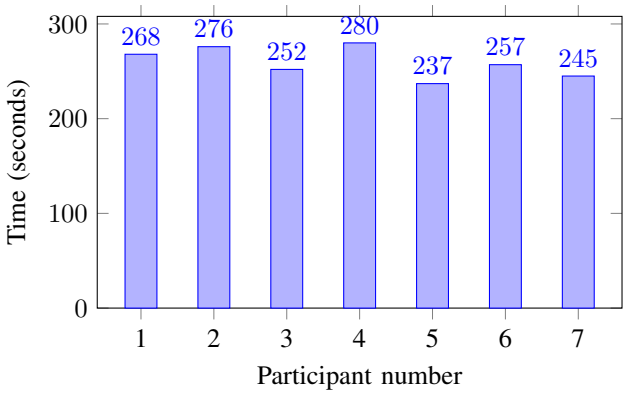


Fig. 4. Execution time of each participant during prototype evaluation.

TABLE I
PSSUQ RESULTS WITH EXPANDED METRICS

| Category | Mean | SD | CV (%) |
|---|---|---|---|
| System Use | 1.09 | 0.14 | 12.8 |
| Information Quality | 2.42 | 1.64 | 67.8 |
| Overall Score | 1.66 | 1.20 | 72.2 |

### *D. Comparative Usability: Experience Level vs. Performance*

The sample was analyzed by grouping participants into three experience ranges (1–2 years, 3–4 years, and 5+ years). Table II summarizes performance and perception. The results indicate that increased experience marginally improved performance, although differences were not statistically significant. This supports the observation that the system has a low dependency on prior AR or PV expertise, reinforcing its accessibility for field use. See the Table II.

TABLE II
COMPARISON BETWEEN PARTICIPANT EXPERIENCE AND PERFORMANCE

| Experience Level | Avg. SysUse Score | Avg. Time (seconds) |
|---|---|---|
| 1–2 years | 1.12 | 264 |
| 3–4 years | 1.08 | 249 |
| 5+ years | 1.05 | 246 |

## V. CONCLUSION

This work presented the development and evaluation of an AR prototype designed to support the accurate placement of PV inverters during site inspections. The prototype integrated BIM, parametric 3D modeling, and marker-based AR visualisation to address recurring operational issues identified in solar engineering workflows. The motivation for this study originated from the observation that inaccurate measurements and incomplete spatial documentation in the early stages of installation were responsible for a measurable portion of engineering rework, additional material costs, and project delays.

The system architecture combined Revit-based parametric modeling with Unity and Vuforia to achieve a mobile application capable of displaying inverter assemblies at real scale. The prototype was validated with seven users who had previous experience in PV deployment. The evaluation included measurement accuracy tests, execution time analysis, and a usability assessment using the Post-Study System Usability Questionnaire (PSSUQ). The results demonstrated that the AR application achieved high spatial fidelity, with an average deviation of approximately 0.5 cm between virtual and real-world measurements. Participants completed the inspection tasks in an average time of 4 minutes and 19 seconds, which suggests that the application requires a minimal learning curve. The PSSUQ results showed strong system usability and positive user perception of information clarity, indicating that the interface and interaction model were effective for field use.

The findings confirm that AR can be used as a practical tool to enhance the accuracy of inverter placement, reduce ambiguity in field documentation, and minimize rework in PV engineering workflows. The prototype also proved to be a valuable resource for communication between field technicians, design teams, and clients, since the real-scale visualization supports a shared understanding of spatial constraints.

Although the system demonstrated strong performance, certain limitations were observed during validation. Some users reported minor inconsistencies in AR anchoring under strong sunlight or windy outdoor conditions, which affected marker stability. The availability of only one inverter model also limited the range of configurations that could be tested. These limitations point to opportunities for system refinement.

Future work may include the expansion of the equipment library to support multiple inverter models, the integration of additional manufacturers' specifications, and the inclusion of advanced AR techniques that do not rely solely on markers. Large-scale field deployment would allow the collection of quantitative data regarding cost reduction and time savings in real installation scenarios. The system may also be adapted for broader applications in electrical maintenance, architectural visualization, and civil engineering inspections, potentially integrating real-time BIM updates through cloud-based platforms.

Overall, the results demonstrate that the proposed AR solution is feasible, effective, and aligned with current trends in digital transformation and Industry 4.0. The prototype provides a structured approach to reducing installation errors and improving decision-making in PV system deployment, supporting the continued professionalization and technological evolution of the solar energy sector.

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